Tutorials for the course BIE-MA1

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Tutorial n. 3

Limits

Limit of a real sequence and of real function; algebraic operations with limits; limit of subequences; Squeeze theorem; Heine's theorem; limit of composite and inverse function; elementary limits; asymptotic hierarchy of infinities.

Notation

e .... Euler's number
ln .... Natural logarithm
$\overline{\mathbb{R}}$ .... extended real line
$\lim\limits_{x\to a} f(x)$ .... limit of the function f at the point a

3.1 On the definition of limit of a sequence

Exercise 3.1: Express into terms of quantifiers and formulas the fact that $\lim_{n\to+\infty} a_n \neq a$, $a \in \mathbb{R}$.

Solution. Recall definition of limit (finite limit case):

Let $a \in \mathbb{R}$

$$\lim_{n\to+\infty} a_n = a$$

if and only if

$$\forall \epsilon > 0 \;\; \exists n_0 \in \mathbb{R} \text{ such that } a_n \in (a-\epsilon, a+\epsilon), \;\; \forall n \geq n_0$$

To solve the exercise we have to write the negation of the above statement. To do it, we replace above the symbols"$\forall$" with "$\exists$" and "$\exists$" with '$\forall$", "$\in$" with "$\notin$" and obtain that $\lim_{n\to+\infty} a_n \neq a$ if and only if

$$\forall n_o \in \mathbb{R}, \;\; \exists \epsilon > 0 \text{ and } \exists n \geq n_0 \text{ such that } a_n \notin (a-\epsilon, a+\epsilon).$$

Exercise 3.2: Which of the following formulas is *equivalent* with the fact that $\lim_{n\to+\infty} a_n \neq a$?

i. There exists a neighborhood $U(a)$ of the point a such that infinitely many members of the sequence (a_n) lie outside $U(a)$.

ii. There exists a neighborhood $U(a)$ of the point a that contains only finitely many members of the sequence (a_n).

iii. There is no neighborhood $U(a)$ of the point a containing infinitely many members of the sequence (a_n).

Statements $i) - ii)$

3.2 Algebraic operations with limits: indeterminate expressions $\frac{\pm\infty}{\pm\infty}$, $\frac{0}{0}$, $\pm\infty \mp \infty$

Exercise 3.3: Find the following limits:

i. $\lim_{n\to+\infty} (5n^3 - 7n + 1)$

ii. $\lim_{n\to+\infty} \dfrac{5n^3 - 7n + 1}{n^2 - 3}$

iii. $\lim_{n\to+\infty} \dfrac{5n^3 - 7n + 1}{n^3 - 3}$

iv. $\lim_{n\to+\infty} \dfrac{5n^3 - 7n + 1}{n^4 - 3}$

Solution.

i. We have the indeterminate expression $+\infty - \infty$. To solve it, we highlight the "biggest infinity", given in this case by the highest power of n

$$\lim_{n\to+\infty} (5n^3 - 7n + 1) = \lim_{n\to+\infty} n^3 \left(5 - \frac{7}{n^2} + \frac{1}{n^3}\right) = +\infty,$$

where we used that $\lim_{n\to+\infty} \frac{7}{n^2} = \lim_{n\to+\infty} \frac{1}{n^3} = 0$ (Remember the rule $\frac{1}{+\infty} = 0$).

ii. We have an ideterminate expression: $\frac{+\infty-\infty}{+\infty}$. To solve the limit, we highlight the "biggest infinity" at both numerator and denominator, by collecting the highest power of n at both numerator and denominator:

$$\lim_{n\to+\infty} \frac{5n^3 - 7n + 1}{n^2 - 3} = \lim_{n\to+\infty} \frac{n^3 \left(5 - \frac{7}{n^2} + \frac{1}{n^3}\right)}{n^2 \left(1 - \frac{3}{n^2}\right)} = +\infty.$$

iii. As above, we have the ideterminate expression: $\frac{+\infty-\infty}{+\infty}$. By collecting the highest power of n at both numerator and denominator we obtain

$$\lim_{n\to+\infty} \frac{5n^3 - 7n + 1}{n^3 - 3} = \lim_{n\to+\infty} \frac{n^3 \left(5 - \frac{7}{n^2} + \frac{1}{n^3}\right)}{n^3 \left(1 - \frac{3}{n^3}\right)} = 5.$$

Note: here we have the quotient of infinities of the same order.

iv. Again, we have the ideterminate expression: $\frac{+\infty-\infty}{+\infty}$. By collecting the highest power of n at both numerator and denominator we obtain

$$\lim_{n\to+\infty} \frac{5n^3 - 7n + 1}{n^4 - 3} = \lim_{n\to+\infty} \frac{n^3 \left(5 - \frac{7}{n^2} + \frac{1}{n^3}\right)}{n^4 \left(1 - \frac{3}{n^4}\right)} = 0.$$

Note: once you will be more familiar with limits, you can also simply write that

$$\lim_{n\to+\infty} \frac{5n^3 - 7n + 1}{n^4 - 3} = \lim_{n\to+\infty} \frac{5n^3}{n^4} = \lim_{n\to+\infty} \frac{5}{n} = 0$$

by recongnizing the "dominant infinity" at numerator and denominator.

Exercise 3.4: Find the limit:

$$\lim_{n\to+\infty} \frac{\frac{1}{n} - \frac{1}{n+4}}{\frac{1}{n+1} - \frac{1}{n+2}}.$$

Solution. Here we have the indeterminate form $\frac{0}{0}$, we can reduce it to $\frac{+\infty}{+\infty}$ as follows:

$$\lim_{n\to+\infty} \frac{\frac{1}{n} - \frac{1}{n+4}}{\frac{1}{n+1} - \frac{1}{n+2}} = 4\frac{(n+1)(n+2)}{n(n+4)} = 4.$$

Exercise 3.5: Find two sequences (a_n) and (b_n) which have the same limit:

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} b_n = +\infty$$

and satisfy one of the following conditions (find an example for each of the conditions):

i. $\lim\limits_{n\to+\infty} (a_n - b_n) = -3$

ii. $\lim\limits_{n\to+\infty} (a_n - b_n) = -\infty$

iii. $\lim\limits_{n\to+\infty} (a_n - b_n) = +\infty$ or explain why it is not possible to find such sequences.

Solution. Here it is one possible solution:

i. $a_n = n,\ b_n = n + 3,$

ii. $a_n = n,\ b_n = n^2,$

iii. $a_n = n^2,\ b_n = n.$

Exercise 3.6: Find two sequences (a_n) and (b_n) such that

$$\lim_{n\to+\infty} a_n = 0, \qquad \lim_{n\to+\infty} b_n = +\infty$$

which also satisfy one of the following conditions (find an example for each of the conditions):

i. $\lim\limits_{n\to+\infty} a_n b_n = 1$

ii. $\lim\limits_{n\to+\infty} a_n b_n = -1$

iii. $\lim\limits_{n\to+\infty} a_n b_n = 0$

iv. $\lim\limits_{n\to+\infty} a_n b_n = -\infty$

or explain why it is not possible to find such sequences.

Solution. Here it is one possible solution:

1. $a_n = \frac{1}{n},\ b_n = n,$
2. $a_n = -\frac{1}{n},\ b_n = n,$
3. $a_n = 0,\ b_n = n,$

4. $a_n = -\frac{1}{n}$, $b_n = n^2$.

Exercise 3.7: Can you find two sequence (a_n) and (b_n) such that $\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} b_n = -\infty$ however $\lim \frac{a_n}{b_n} = 2$.? If yes, give an example, if not explain why it is not possible.

Exercise 3.8: Can you find two sequences (a_n) and (b_n) such that $\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} b_n = 0$, however $\lim \frac{a_n}{b_n} = -\infty$? If yes, give an example, if not explain why it is not possible.

Exercise 3.9: Calculate the following limit:

$$\lim_{n\to+\infty} \sqrt{2n+1} - \sqrt{n}.$$

Solution. Here we have the indeterminate expression $+\infty - \infty$. To solve it, we can proceed as follows

$$\begin{aligned} \lim_{n\to+\infty} \sqrt{2n+1} - \sqrt{n} &= \lim_{n\to+\infty} (\sqrt{2n+1} - \sqrt{n}) \frac{\sqrt{2n+1} + \sqrt{n}}{\sqrt{2n+1} + \sqrt{n}} \\ &= \lim_{n\to+\infty} \frac{2n+1-n}{\sqrt{2n+1} + \sqrt{n}} \\ &= \lim_{n\to+\infty} \frac{n+1}{\sqrt{2n+1} + \sqrt{n}}. \end{aligned} \tag{3.1}$$

We have reduced in this way to the indeterminate expression $\frac{+\infty}{+\infty}$, that we can solve as in the previous exercises:

$$\lim_{n\to+\infty} \frac{n+1}{\sqrt{2n+1} + \sqrt{n}} = \lim_{n\to+\infty} \frac{n\left(1+\frac{1}{n}\right)}{\sqrt{n}\left(\sqrt{(2+\frac{1}{n})}+1\right)} = +\infty.$$

Exercise 3.10: Find the limit:

$$\lim_{n\to+\infty} n\left(\sqrt{n^2+1} - n\right).$$

Solution.

$$\lim_{n\to+\infty} n\left(\sqrt{n^2+1} - n\right) = \lim_{n\to\infty} n\frac{1}{\sqrt{n^2+1}+n} = \frac{1}{2}.$$

Exercise 3.11: Find the limit:

$$\lim_{n\to+\infty} \frac{2^n - 2^{-n}}{2^n + 2^{-n}}$$

Solution.

$$\lim_{n\to+\infty} \frac{2^n - 2^{-n}}{2^n + 2^{-n}} = 1.$$

(Remember that $\lim_{n\to+\infty} 2^{-n} = 0$).

Exercise 3.12: Compute the limit:

$$\lim_{n\to+\infty} \frac{\left(\frac{1}{2}\right)^{n^2} - \left(\frac{1}{3}\right)^{n^2}}{\left(\frac{1}{2}\right)^{n^2+1} - \left(\frac{1}{3}\right)^{n^2+1}}.$$

Recall

$$\lim_{n\to+\infty} a^n = \begin{cases} 0, & |a|<1 \\ 1, & a=1 \\ +\infty, & a>1 \\ \nexists, & a\le -1 \end{cases}$$

Solution. We have the indeterminate expression $\frac{0}{0}$. To solve it, we highlight the "slowest" term that reaches zero at both numerator and denominator:

$$\lim_{n\to+\infty} \frac{\left(\frac{1}{2}\right)^{n^2} - \left(\frac{1}{3}\right)^{n^2}}{\left(\frac{1}{2}\right)^{n^2+1} - \left(\frac{1}{3}\right)^{n^2+1}} = \lim_{n\to+\infty} \frac{(\frac{1}{2})^{n^2}(1-(\frac{2}{3})^{n^2})}{(\frac{1}{2})^{n^2+1}(1-(\frac{2}{3})^{n^2+1})} = 2.$$

Exercise 3.13: Find the following limits of function:

$$a)\ \lim_{x\to0} \frac{(1+x)(1+2x)(1+3x)-1}{x}, \quad b)\ \lim_{x\to0} \frac{-3x^3+x^2+x}{x^4}$$

Solution:a) 6, b) the limit does not exist

Solution. a) We have the indeterminate expression $\frac{0}{0}$. We can solve it as follows

$$\begin{aligned} \lim_{x\to0} \frac{(1+x)(1+2x)(1+3x)-1}{x} &= \lim_{x\to0} \frac{(1+3x+2x^2)(1+3x)-1}{x} \\ &= \lim_{x\to0} \frac{11x^2+6x^3+6x}{x} \\ &= \lim_{x\to0} \frac{x(6x^2+11x+6)}{x} = \qquad (3.2) \\ &= \lim_{x\to0} 6x^2+11x+6 = 0+0+6 = 6 \end{aligned}$$

where in (3.2) we collected at numerator the "slowest zero", i.e. the slowest term that reaches zero, given by the lowest power of x.

Exercise 3.14: Find the limits of function:

$$a)\ \lim_{x\to2} \frac{4-x^2}{x-2}, \quad b)\ \lim_{x\to-1} \frac{x^3-2x-1}{x^5-2x-1}.$$

Solution: a) -4 b) $\frac{1}{3}$

Solution. b) We have the indeterminate expression $\frac{0}{0}$.
Note Here x is not going to zero, nor to infinity. Thus, be careful: collecting the lowest/highest power of x at numerator and denominator does not help in solving the limit!

In this case $x \to -1$ and -1 is a root of both polynomials at numerator and denominator. To solve the indeterminate expression we have to factorize $(x+1)$ out (this is the quantity that goes to zero!) at both numerator and denominator.

Recall: The polynomial long division. Let us divide x^3-2x-1 by $x+1$:

$$
\begin{array}{r|l}
x^3+0x^2-2x-1 & x+1 \\ \hline
-x^3-\ x^2 \qquad\qquad & x^2-x-1 \longleftarrow \text{Quotient} \\
-x^2-2x \quad & \\
x^2+\ x \quad & \\
-x-1 & \\
x+1 & \\
\text{Reminder}\longrightarrow \qquad 0 &
\end{array}
$$

We found that

$$(x^3-2x-1):(x+1)=x^2+x-1.$$

Thus, we can factorize x^3-2x-1 as

$$x^3-2x-1=(x+1)(x^2-x-1) \tag{3.3}$$

Similarly, we can find (try!)

$$(x^5-2x-1):(x+1)=x^4-x^3+x^2-x-1,$$

therefore

$$x^5-2x-1=(x+1)(x^4-x^3+x^2-x-1). \tag{3.4}$$

By using (3.3) and (3.4), the limit simplifies to

$$
\begin{aligned}
\lim_{x\to-1}\frac{x^3-2x-1}{x^5-2x-1} &= \lim_{x\to-1}\frac{(x+1)(x^2-x-1)}{(x+1)(x^4-x^3+x^2-x-1)} \\
&= \lim_{x\to-1}\frac{x^2-x-1}{x^4-x^3+x^2-x-1}=\frac{1+1-1}{1+1+1+1-1}=\frac{1}{3}
\end{aligned} \tag{3.5}
$$

3.3 Theorems on the limits of sequences: squeeze theorem and theorem on the limit of a subsequence

Exercise 3.15: Do the following limits of a sequence exist? If yes, find the value.

a) $\displaystyle\lim_{n\to+\infty}\frac{\sin(n)}{n}$,

b) $\displaystyle\lim_{n\to+\infty}\left(1+(-1)^n\right)n$,

c) $\displaystyle\lim_{n\to+\infty}\frac{\frac{1}{2^n}+\frac{1}{3^n}}{\frac{2}{3^n}+\frac{1}{2^n}}$.

d) $\displaystyle\lim_{n\to+\infty}\left(\frac{1}{2}+3(-1)^n\right)n$

a) 0, b) it does not exist, c) 1, d) it does not exist,

Solution. a) Let $b_n=\frac{\sin(n)}{n}$. By increasing n, the sequence $\sin(n)$ takes values between -1 and 1 repeatedly, this might suggest that b_n does not have limit. On the other hand $\sin(n)$ remains a finite quantity for $n\to+\infty$, since it anyway takes values in $[-1,1]$. Thus in the limit for $n\to+\infty$ we would have a finite quantity (but with values oscillating in $[-1,1]$) divided by $+\infty$... does b_n have limit or not?

By using the idea that anyway $-1 \leq \sin(n) \leq 1$, we apply here the Squeeze Theorem: we look for two sequences a_n and c_n that have the same limit α and such that

$$a_n \leq \frac{\sin(n)}{n} \leq c_n$$

Then, by the squeeze theorem, we can conclude that $\lim\limits_{n\to+\infty} \frac{\sin(n)}{n} = \alpha$. We can take $a_n = -\frac{1}{n}$ and $c_n = \frac{1}{n}$ and have

$$-\frac{1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}.$$

Since $\lim\limits_{n\to+\infty} (-\frac{1}{n}) = \lim\limits_{n\to+\infty} \frac{1}{n} = 0$, then by the squeeze thm.,

$$\lim_{n\to+\infty} \frac{\sin(n)}{n} = 0.$$

Exercise 3.16: Find the following limits of sequence, or if the limit does not exist, explain why.

a)

$$\lim_{n\to+\infty} \frac{\cos n}{n}$$

(hint: use $|\cos x| < 1$ and the squeeze thm.)

b)

$$\lim_{n\to+\infty} \tan\frac{1}{n}$$

(hint: use $\cos x > \frac{1}{2}$ for $|x| < \frac{\pi}{3}$).

Exercise 3.17: Find the following limits of sequence, or if the limit does not exist, explain why.

$$\lim_{n\to+\infty} \frac{2-\cos(n^2)}{\sqrt{n}}$$

Solution:0

Exercise 3.18: Find the following limits of sequence, or if the limit does not exist, explain why.

$$\lim_{n\to+\infty} \frac{\sqrt{n}+\sin(2n^3)}{n^2}$$

Solution: 0

Exercise 3.19: Find the following limits of sequence, or if the limit does not exist, explain why.

$$\lim_{n\to+\infty} n^2 - \arcsin\left(\frac{1}{n}\right)$$

(hint: use $|\arcsin\left(\frac{1}{n}\right)| \leq \frac{\pi}{2}$ and the squeeze thm.) Solution: $+\infty$

Exercise 3.20: Find the following limits of sequence, or if the limit does not exist, explain why.

$$\lim_{n\to+\infty} \frac{\arccos\left(\frac{1}{n}\right) - 3n^2}{n}$$

(hint: use $0 \le \arccos\left(\frac{1}{n}\right) \le \pi$ and the squeeze thm.) Solution: $-\infty$

Exercise 3.21: Find the following limits of sequence, or if the limit does not exist, explain why.

a)

$$\lim_{n\to+\infty} n \sin\frac{2}{n} \cos\frac{2}{n}$$

b)

$$\lim_{n\to+\infty} n^2 \left(1 - \cos\frac{1}{n}\right)$$

c)

$$\lim_{n\to+\infty} \frac{n\left(1 - \cos\frac{1}{n}\right)}{\sin\frac{1}{n}}$$

d)

$$\lim_{n\to+\infty} \frac{n^2+1}{n+1} \sin\frac{1}{n}$$

a) 2, b) $\frac{1}{2}$, c) $\frac{1}{2}$, d) 1 .

Exercise 3.22: Find the following limit of a sequence

$$\lim_{n\to+\infty} \sqrt[n]{4n^3+5}.$$

Hint: Find suitable upper and lower bound of the sequence and use the fact that $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
Solution: 1

Solution. $\forall\, n \in \mathbb{N}$ it holds

$$1 \le \sqrt[n]{n} \le \sqrt[n]{4n^3+5} \le \sqrt[n]{9n^3} = \sqrt[n]{9}\,\sqrt[n]{n}^3.$$

Since $\sqrt[n]{n}^3 \to 1$, by applying the Squeeze Thm. we conclude that

$$\lim_{n\to+\infty} \sqrt[n]{4n^3+5} = 1.$$

Exercise 3.23: Find the following limit of a sequence

$$\lim_{n\to+\infty} \frac{\lfloor\sqrt{n}\rfloor}{\sqrt{n}}.$$

Recall: $\lfloor x\rfloor$ denotes the lower integer part of real number x and by its definition it satisfies $\lfloor x\rfloor - 1 < \lfloor x\rfloor \le x$.

Solution. $\forall\, n \in \mathbb{N}$ we have

$$\frac{\sqrt{n}-1}{\sqrt{n}} < \frac{\lfloor\sqrt{n}\rfloor}{\sqrt{n}} \le \frac{\sqrt{n}}{\sqrt{n}} = 1.$$

Since $\frac{\sqrt{n}-1}{\sqrt{n}} = 1$ (details of the computation are left to you!), by the squeeze thm,

$$\lim_{n\to+\infty} \frac{\lfloor \sqrt{n} \rfloor}{\sqrt{n}} = 1.$$

Exercise 3.24: Determine the following limit of a sequence:

$$\lim_{n\to+\infty} (-1)^{\frac{n(n+1)}{2}} \frac{n}{n+1}.$$

The limit does not exist (why?)

Solution. Let $a_n = (-1)^{\frac{n(n+1)}{2}} \frac{n}{n+1}$. Let us notice that $\lim_{n\to+\infty} \frac{n}{n+1} = 1$. However, for $\frac{n(n+1)}{2} = 2\ell$ we have $(-1)^{\frac{n(n+1)}{2}} = 1$ and for $\frac{n(n+1)}{2} = 2\ell+1$ we have $(-1)^{\frac{n(n+1)}{2}} = -1$, $\ell \in \mathbb{N}$. This suggests that the limit might not exists. To prove it rigorously, we look for two subsequences that have different limits. One possible choice is

$$a_{n_k} = a_{4k} = (-1)^{2k(4k+1)} \frac{4k}{4k+1}, \quad \text{and} \quad a_{\ell_k} = a_{4k+2} = (-1)^{(2k+1)(4k+3)} \frac{4k+2}{4k+3}.$$

We have

$$\lim_{k\to+\infty} a_{n_k} = +\infty, \quad \text{and} \quad \lim_{k\to+\infty} a_{\ell_k} = -\infty,$$

thus the sequence (a_n) has no limit.

3.4 Theorems on limits of functions: Squeeze theorem for functions and Heine's theorem

Exercise 3.25: Find the following limit of function

$$\lim_{x\to+\infty} \frac{x - 3\arctan\left(\frac{1}{x}\right)}{2x-3}.$$

Solution. The function is well defined on a neighborhood of $+\infty$, namely on $(\frac{3}{2}, +\infty)$. In addition, the denominator is positive on this interval and

$$\frac{x - 3\cdot\frac{\pi}{2}}{2x-3} \le \frac{x - 3\arctan(\frac{1}{x})}{2x-3} \le \frac{x + 3\cdot\frac{\pi}{2}}{2x-3}.$$

It is not difficult to show that the functions most on the left and most on the right have the same limit equal to $\frac{1}{2}$ (but do it!). By the squeeze theorem, the function in the middle converges to $\frac{1}{2}$, too.

Exercise 3.26: Does the following limit of a function exists?

$$\lim_{x\to 0} \frac{1}{x} \sin\left(\frac{1}{x}\right).$$

The limit does not exist. To prove it, use Heine's thm and two suitable subsequences....

Exercise 3.27: Find the following limit of function:

$$\lim_{x\to 0} \cos\frac{1}{x},$$

or if the limit does not exist, explain why.

Exercise 3.28: Find the following limit of function:

$$\lim_{x\to 0} x\cos\frac{1}{x},$$

or if the limit does not exist, explain why.

3.5 Limit of composition of functions and limit of inverse function

Exercise 3.29: Find the following limit of a sequence:

$$\lim_{n\to+\infty}\frac{\ln\left(n^2+4n+2\right)}{\ln\left(3n^4+5\right)}$$

Solution: $\frac{1}{2}$

Exercise 3.30: Compute the following limit of sequence (if the limit does not exist, explain why):

$$\lim_{n\to+\infty}\frac{\ln(n^3-3n+2)}{\ln(-n^2+3n^6)}.$$

$\frac{1}{2}$

Exercise 3.31: Find the following one-sided limits of function:

$$a)\lim_{x\to 1^+}\operatorname{arctg}\frac{1}{1-x},\quad b)\lim_{x\to 1^-}\operatorname{arctg}\frac{1}{1-x}.$$

Solution: a) $-\frac{\pi}{2}$, b) $\frac{\pi}{2}$

Solution. a) Since

$$\lim_{x\to 1^+}\frac{1}{1-x}=\frac{1}{0^-}=-\infty,$$

we have

$$\lim_{x\to 1^+}\arctan\frac{1}{1-x}=-\frac{\pi}{2}.$$

Solution. b) Since

$$\lim_{x\to 1^-}\frac{1}{1-x}=\frac{1}{0^+}=+\infty,$$

we have

$$\lim_{x\to 1^-}\arctan\frac{1}{1-x}=\frac{\pi}{2}$$

Exercise 3.32: Find the following limits of function:

$$a)\lim_{x->+\infty}\frac{x}{\sqrt{1+x^2}},\quad b)\lim_{x\to-\infty}\frac{x}{\sqrt{1+x^2}}.$$

Solution: a) 1, b) -1

Solution. a) We have the indeterminate expression $\frac{+\infty}{+\infty}$. By collecting the highest power of x at both numerator and denominator we simplify to

$$\begin{aligned} \lim_{x->+\infty} \frac{x}{\sqrt{1+x^2}} &= \lim_{x->+\infty} \frac{x}{\sqrt{x^2\left(1+\frac{1}{x^2}\right)}} \\ & \lim_{x->+\infty} \frac{x}{|x|\sqrt{\left(1+\frac{1}{x^2}\right)}} \\ & \lim_{x->+\infty} \frac{x}{|x|} = \lim_{x->+\infty} \frac{x}{x} = 1. \end{aligned} \tag{3.6}$$

b) We have the indeterminate expression $\frac{+\infty}{+\infty}$ as above. Similarly, by collecting the highest power of x at both numerator and denominator we simplify to

$$\begin{aligned} \lim_{x->-\infty} \frac{x}{\sqrt{1+x^2}} &= \lim_{x->-\infty} \frac{x}{\sqrt{x^2\left(1+\frac{1}{x^2}\right)}} \\ & \lim_{x->-\infty} \frac{x}{|x|\sqrt{\left(1+\frac{1}{x^2}\right)}} \\ & \lim_{x->-\infty} \frac{x}{|x|} = \lim_{x->-\infty} \frac{x}{-x} = -1. \end{aligned} \tag{3.7}$$

Note: for values of x close to $-\infty$, $\sqrt{x^2} = |x| = -x$!

Exercise 3.33: Find the following limit of function:

$$a) \lim_{x->-\infty} \arcsin \frac{x}{\sqrt{1+x^2}}, \qquad b) \lim_{x\to+\infty} \arcsin \frac{x}{\sqrt{1+x^2}}.$$

Solution: a) $-\frac{\pi}{2}$, b) $\frac{\pi}{2}$

Solution. From the previous exercise we deduce that

$$\lim_{x->-\infty} \arcsin \frac{x}{\sqrt{1+x^2}} = \arcsin(-1) = -\frac{\pi}{2}$$

and

$$\lim_{x->+\infty} \arcsin \frac{x}{\sqrt{1+x^2}} = \arcsin(1) = \frac{\pi}{2}.$$

3.6 Application of elementary limits

Do not use l'Hospital's rule in the following exercises. You can *sometimes* use the following elementary limits instead:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \qquad \lim_{x\to 0} \frac{\ln(1+x)}{x} = 1 \qquad \lim_{x\to 0} \frac{e^x - 1}{x} = 1 \qquad \lim_{x\to+\infty} \left(1+\frac{1}{x}\right)^x = e.$$

Exercise 3.34: Find the following limit of function:

$$\lim_{x\to 0} \frac{\sin(5x)}{x}.$$

Solution: 5

Exercise 3.35: Find the following limit of function:

$$\lim_{x\to 0} \frac{\sin(2x)}{\sin(3x)}.$$

Solution: $\frac{2}{3}$

Solution. When $x \to 0$ both numerator and denominator goes to zero, thus we have the indeterminate expression $\frac{0}{0}$. To solve it, we can use, after some simplifications, the rule

$$\lim_{z\to 0} \frac{\sin z}{z} = 1$$

that implies that also

$$\lim_{z\to 0} \frac{z}{\sin z} = 1.$$

First of all, we can write the limit as

$$\lim_{x\to 0} \frac{\sin(2x)}{2x} \frac{3x}{\sin(3x)} \frac{2x}{3x}$$

Let us now focus on

$$\lim_{x\to 0} \frac{\sin(2x)}{2x}$$

we can set $z = 2x \to 0$, thus we can apply the rule

$$\lim_{x\to 0} \frac{\sin(2x)}{2x} = \lim_{z\to 0} \frac{\sin z}{z} = 1.$$

Similarly, we have that

$$\lim_{x\to 0} \frac{3x}{\sin(3x)} = 1$$

Thus,

$$\lim_{x\to 0} \frac{\sin(2x)}{\sin(3x)} = \lim_{x\to 0} \frac{\sin(2x)}{2x} \frac{3x}{\sin(3x)} \frac{2x}{3x} = 1 \cdot 1 \cdot \frac{2}{3} = \frac{2}{3}.$$

Exercise 3.36: Find the following limits of function:

$$a)\ \lim_{x\to +\infty} \frac{\sin x^2}{\sqrt{x}}, \quad b)\ \lim_{x\to -\infty} \frac{\sin x^3}{x^2}.$$

Solution: a) 0, b) 0

Solution. Let us solve here exercise b). The solution for a) is similar. We can use the Squeeze theorem. We have

$$-\frac{1}{x}^2 \le \frac{\sin x^3}{x^2} \le \frac{1}{x}^2$$

Since

$$\lim_{x\to -\infty} \frac{-1}{x^2} = \lim_{x\to -\infty} \frac{1}{x^2} = 0,$$

by the Squeeze theorem, also

$$\lim_{x\to -\infty} \frac{\sin x^3}{x^2} = 0.$$

Exercise 3.37: Find the following limit of function:

$$\lim_{x\to 0} \frac{1-\cos x}{x^2}.$$

Solution: $\frac{1}{2}$

Solution. Here we have the indeterminate expression $\frac{0}{0}$. We can solve it the following way:

$$\begin{aligned}
\lim_{x\to 0} \frac{1-\cos x}{x^2} &= \lim_{x\to 0} \frac{1-\cos x}{x^2}\frac{1+\cos x}{1+\cos x} \\
&= \lim_{x\to 0} \frac{1-\cos^2 x}{x^2}\frac{1}{1+\cos x} \\
&= \lim_{x\to 0} \frac{\sin^2 x}{x^2}\frac{1}{1+\cos x} = 1\cdot\frac{1}{2} = \frac{1}{2},
\end{aligned} \tag{3.8}$$

where in the last line we used the fundamental identity of trigonometry:

$$\sin^2 x + \cos^2 x = 1$$

and the fact that

$$\lim_{x\to 0} \frac{\sin^2 x}{x^2} = \lim_{x\to 0} \frac{\sin x}{x}\frac{\sin x}{x} = \lim_{x\to 0} \frac{\sin x}{x} \lim_{x\to 0} \frac{\sin x}{x} = 1\cdot 1 = 1.$$

Exercise 3.38: Find the following limits of function:

$$a)\ \lim_{x\to+\infty} e^x, \quad b)\ \lim_{x\to-\infty} e^x.$$

Solution: a) $+\infty$, b) 0

Exercise 3.39: Find the following limit of function:

$$\lim_{x\to+\infty} \frac{\ln(1+e^x)}{x}$$

Solution: 1

<u>Note</u> In this and in the following exercises, do not get confused with the rule

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1!$$

In this exercise the limit is not for $x \to 0$!

Solution. We have the indeterminate expression $\frac{+\infty}{+\infty}$. We can solve it as follows:

$$\begin{aligned}
\lim_{x\to+\infty} \frac{\ln(1+e^x)}{x} &= \lim_{x\to+\infty} \frac{\ln\left(e^x\left(\frac{1}{e^x}+1\right)\right)}{x} \\
&= \lim_{x\to+\infty} \frac{\ln e^x + \ln\left(\frac{1}{e^x}+1\right)}{x} \\
&= \lim_{x\to+\infty} \frac{\ln e^x}{x} = \lim_{x\to+\infty} \frac{x\ln e}{x} = \lim_{x\to+\infty} \frac{x}{x} = 1.
\end{aligned} \tag{3.9}$$

Exercise 3.40: Find the following limit of function:

$$\lim_{x\to-\infty} \frac{\ln(1+e^x)}{e^x}.$$

Solution: 1

Solution. Remember that $\ln(1) = 0$ and $\lim_{x\to-\infty} e^x = 0!$. Thus, we have

$$\lim_{x\to-\infty} \frac{\ln(1+e^x)}{x} = \frac{0}{-\infty} = 0$$

Exercise 3.41: Find the following limit of function:

$$\lim_{x\to 0} \frac{e^{3x}-1}{x}.$$

Solution: 3

Exercise 3.42: Find the following limit of function:

$$\lim_{x\to 0} \frac{e^{3x}-e^{4x}}{x}.$$

Solution: -1

Solution. Here we have the indeterminate expression $\frac{0}{0}$. To solve it, we can use the rule

$$\lim_{z\to 0} \frac{e^z-1}{z} = 1.$$

In fact, we have

$$\begin{aligned}
\lim_{x\to 0} \frac{e^{3x}-e^{4x}}{x} &= \lim_{x\to 0} \frac{e^{3x}-1+1-e^{4x}}{x} \\
&= \lim_{x\to 0} \left(\frac{e^{3x}-1}{x} + \frac{1-e^{4x}}{x} \right) \\
&= \lim_{x\to 0} \left(3\frac{e^{3x}-1}{3x} - 4\frac{e^{4x}-1}{4x} \right) \\
&= 3\lim_{x\to 0} \frac{e^{3x}-1}{3x} - 4\lim_{x\to 0} \frac{e^{4x}-1}{4x} \\
&= 3\lim_{z\to 0} \frac{e^{z}-1}{z} - 4\lim_{w\to 0} \frac{e^{w}-1}{w} = 3\cdot 1 - 4\cdot 1 = -1,
\end{aligned}$$

where in the last line we set $z = 3x$ and $w = 4x$ and use the fact that $z \to 0$ and $w \to 0$ for $x \to 0$.

3.7 Asymptotic hierarchy of infinities

Exercise 3.43: Find the following limits of sequence (if the limit does not exist, explain why):

a) $\lim_{n\to+\infty} \frac{n}{2^n}$,

b) $\lim\limits_{n\to+\infty} \dfrac{n^2+1}{e^n+n^3}$

Exercise 3.44: Compute the following limits of a sequence (if the limit does not exist, explain why):

a)

$$\lim_{n\to+\infty} \frac{-4^{-n}+\sin(n^{14})+n^n}{\sqrt[5]{n}+5^n}.$$

b)

$$\lim_{n\to+\infty} \frac{\sqrt[3]{n}-3^n}{\cos(n^8)+2^{-n}+n!}.$$

Solution: a) $+\infty$, b) 0.

Recall Asymptotic hierarchy of infinities (for sequences):

$$\log_c(n) \ll n^b \ll a^n \ll n! \ll n^n, \quad a,b,c \in \mathbb{R},\ c>1,\ a>1,\ b>0.$$

In terms of limits,

$$\lim_{n\to+\infty} \frac{\log_c n}{n^b} = \lim_{n\to+\infty} \frac{n^b}{a^n} = \lim_{n\to+\infty} \frac{a^n}{n!} = \lim_{n\to+\infty} \frac{n!}{n^n} = 0.$$

Solution. a) We have to find

$$\lim_{n\to+\infty} \frac{-4^{-n}+\sin(n^{14})+n^n}{\sqrt[5]{n}+5^n}.$$

At numerator we have

$$\lim_{n\to+\infty} (-4^{-n}+\sin(n^{14})+n^n) = \lim_{n\to+\infty} n^n = +\infty.$$

At denominator as well

$$\lim_{n\to+\infty} \sqrt[5]{n}+5^n = +\infty.$$

Thus, we have the indeterminate expression $\frac{+\infty}{+\infty}$. To solve it, we highlight the "fastest infinity" at both numerator and denominator. At numerator there is only one term going to infinity in the limit, that is n^n ($4^{-n} \to 0$ and $\sin(n^{14})$ stays bounded in the limit). At denominator, by the asymptotic hierarchy of infinities we see that the dominant infinity in the limit is given by 5^n. Therefore

$$\lim_{n\to+\infty} \frac{-4^{-n}+\sin(n^{14})+n^n}{\sqrt[5]{n}+5^n} = \lim_{n\to+\infty} \frac{n^n}{\sqrt[5]{n}+5^n} = \lim_{n\to+\infty} \frac{n^n}{5^n\left(\frac{\sqrt[5]{n}}{5^n}+1\right)} = \lim_{n\to+\infty} \frac{n^n}{5^n} = +\infty,$$

where by the asymptotic hierarchy of infinities

$$\lim_{n\to+\infty} \frac{\sqrt[5]{n}}{5^n} = 0, \quad \lim_{n\to+\infty} \frac{n^n}{5^n} = +\infty$$

Note Once more familiar with limits, you can skip the last but one equality above, and simply write that by the asymptotic hierarchy of infinities

$$\lim_{n\to+\infty} \lim_{n\to+\infty} \frac{n^n}{\sqrt[5]{n}+5^n} = \lim_{n\to+\infty} \frac{n^n}{5^n} = +\infty.$$

Exercise 3.45: Compute the following limit of a sequence (if the limit does not exist, explain why):

$$\lim_{n\to+\infty} \frac{\sqrt{n^3} - 3^{-n}}{\cos(n^6) + 3^n + 2n!}.$$

3.8 Miscellanea

Exercise 3.46: Find the following limit of a sequence

$$\lim_{n\to+\infty} \frac{\ln(3^n+5)}{\ln(4^n-2)}.$$

Solution. We have the indeterminate expression $\frac{+\infty}{+\infty}$. We can solve it by collecting the dominant infinite inside the arguments of the logarithms and then by applying the properties of the logarithm:

$$\lim_{n\to+\infty} \frac{\ln(3^n+5)}{\ln(4^n-2)} = \lim_{n\to+\infty} \frac{\ln(3^n(1+\frac{5}{3^n}))}{\ln(4^n(1-\frac{2}{4^n}))} = \tag{3.10}$$

$$= \lim_{n\to+\infty} \frac{\ln 3^n + \ln(1+\frac{5}{3^n})}{\ln 4^n + (1-\frac{2}{4^n})} = \tag{3.11}$$

$$= \lim_{n\to+\infty} \frac{n\ln 3}{n\ln 4} = \frac{\ln 3}{\ln 4}. \tag{3.12}$$

<u>Note</u>: we also used the thm. on the limit of composition of functions here.

Exercise 3.47: Compute the limits

a) $\displaystyle\lim_{n\to+\infty} \frac{(n+1)^3-(n-1)^3}{(n+1)^2+(n-1)^2}$,

b) $\displaystyle\lim_{n\to+\infty} \frac{\sqrt[3]{n^3+n}}{n+1}$,

c) $\displaystyle\lim_{n\to+\infty} \frac{n^2+2}{n+4} - \frac{1+2n}{2+\frac{1}{n}}$,

a) 3, b) 1, c) -4

Exercise 3.48: Find the following limits of function, or if the limit does not exist, explain why

a) $\displaystyle\lim_{x\to+\infty} \left(1+\frac{1}{3x^2}\right)^x$

b) $\displaystyle\lim_{x\to 0} (1+2x)^{\frac{3}{x}}$

c) $\displaystyle\lim_{x\to-\infty} \left(1-\frac{3}{x}\right)^{x^3}$

d) $\displaystyle\lim_{x\to-\infty} \left(1+\frac{2}{x}\right)^{x^2}$

a) 1, b) $e^{\frac{3}{2}}$, c) 0, d) 0

Exercise 3.49: Find the following limits of function

a) $\lim\limits_{x\to 2} \dfrac{x^3+2x^2-8x}{x^3-3x^2-4x+12}$,

b) $\lim\limits_{x\to 2+} \dfrac{x^2+3x-4}{x^3-4x^2+x+6}$,

c) $\lim\limits_{x\to 0} \dfrac{x}{e^x-e^{-x}}$,

d) $\lim\limits_{x\to 1} \dfrac{e^x-e}{x-1}$.

a) -3, b) $-\infty$, c) $\frac{1}{2}$, d) e.

Exercise 3.50: Find the following limits of function

a) $\lim\limits_{x\to 0} \dfrac{\cos(x)}{x}\left(2\sin(x)+x\right)$,

b) $\lim\limits_{x\to \pi/2} \dfrac{\cos(x)}{x-\frac{\pi}{2}}$,

c) $\lim\limits_{x\to 0} \dfrac{\operatorname{tg}(x)}{\operatorname{tg}(2x)}$,

d) $\lim\limits_{x\to 0} \dfrac{e^{2x}-1}{\sin(x)}$.

a) 3, b) -1, c) $\frac{1}{2}$, d) 2.

Exercise 3.51: Find the limits of sequence (if it exists):

a) $\lim\limits_{n\to +\infty} \dfrac{100n}{n^2+1}$,

b) $\lim\limits_{n\to +\infty} \dfrac{\sqrt[3]{n^2}\sin n!}{n+1}$,

Exercise 3.52: Find the following limits of function

a) $\lim\limits_{x\to +\infty} \dfrac{x^4-5x}{x^2-3x+1}$,

b) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x^2}-1}{x}$

c) $\lim\limits_{x\to +\infty} \left(\dfrac{x^3}{x^2+1}-x\right)$,

a) $+\infty$, b) 0, c) 0.

Exercise 3.53: Find the following limits of function

a) $\lim\limits_{x\to 0} \left(\dfrac{1}{\sin(x)}-\dfrac{1}{\operatorname{tg}(x)}\right)$,

b) $\lim\limits_{x\to\pi/4} \dfrac{\cos x - \sin x}{\cos(2x)}$,

c) $\lim\limits_{x\to e} \dfrac{\ln(x) - 1}{x - e}$,

d) $\lim\limits_{x\to 0} \dfrac{e^{x^2} - \cos x}{x^2}$.

a) 0, b) $\frac{\sqrt{2}}{2}$, c) $\frac{1}{e}$, d) $\frac{3}{2}$.

Exercise 3.54: Find the following limits of sequence (or if the limit does not exist, explain why):

a) $\lim\limits_{n\to+\infty} n \sin\left(\dfrac{1}{n^2}\right)$

b) $\lim\limits_{n\to+\infty} n^4(1 - e^{\frac{1}{n^3}})$

a) 0, b) ∞.

Exercise 3.55: Find the following limits of function (or if the limit does not exist, explain why):

a) $\lim\limits_{x\to+2} \dfrac{x^3 + 6x^2 - 12x - 8}{x^3 - 8}$

b) $\lim\limits_{x\to+3} \dfrac{x^4 - 81}{x^2 - 6x + 9}$

a) 0, b) the limit does not exist.

Exercise 3.56: Find the following limits of function

a) $\lim\limits_{x\to+\infty} \dfrac{x^2 + 2x^x + \log(x)}{3x^3 + e^{-x^2}}$,

b) $\lim\limits_{x\to+\infty} \dfrac{\ln(x^{10}) + 3^{x+1}}{3x^3 + 3^x}$

c) $\lim\limits_{x\to-\infty} \dfrac{3x^4 + \ln|x|}{x^2 + 5^{-x^3}}$

Solution: a) $+\infty$, b) 3, c) 0